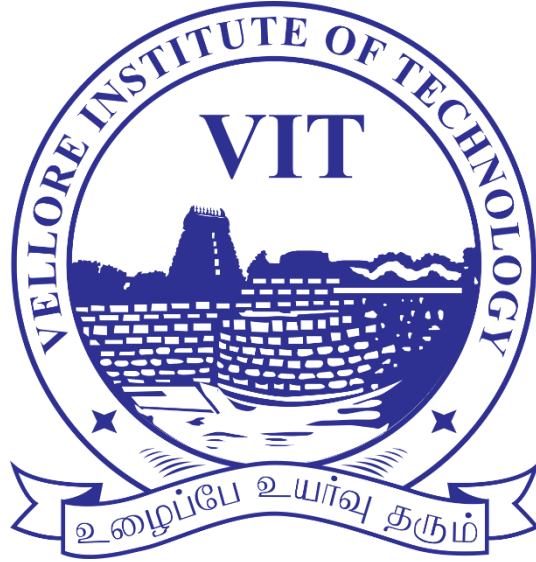




Digital Assignment – 1

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Course Name: MDI3003 – Advanced Predictive Analytics (Lab)

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House Price Prediction using Linear Regression

1) Executive Summary

Accurate house price prediction plays an important role in helping buyers, sellers, banks, investors, and government agencies make better decisions. Traditional valuation methods often depend on manual assessment and expert opinion, which can be time consuming and may produce inconsistent results. This project explores the use of machine learning to predict residential property prices using different characteristics of a house. The study uses the California Housing and Ames Housing datasets to build and compare several regression models. The overall workflow includes data preprocessing, exploratory data analysis, feature engineering, model training, hyperparameter tuning, and performance evaluation. A Simple Linear Regression model is developed as the baseline, followed by Multiple Linear Regression, Ridge Regression, Lasso Regression, Decision Tree, Random Forest, and Gradient Boosting Regression.

The models are evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score to determine the most accurate and reliable approach. The final model is selected based on its overall predictive performance, generalization ability, and robustness across the evaluation metrics. The selected model demonstrates a noticeable improvement over the baseline model, providing more accurate and consistent house price predictions. The analysis shows that features such as living area, neighborhood, overall quality, and median income have the greatest influence on house prices. The model performs well for most properties but may produce

larger prediction errors for luxury homes or properties that are underrepresented in the dataset. Overall, the developed framework serves as a reliable decision support tool for residential property valuation while maintaining a good balance between prediction accuracy and model interpretability.

2) Business Problem and Scope

Accurate property valuation is important for buyers, sellers, banks, investors, and government agencies, as it directly influences financial decisions. Traditional valuation methods often rely on manual assessment and expert judgment, which can be time consuming and may produce inconsistent results. This project addresses this challenge by developing a machine learning model that predicts residential property prices using features such as location, living area, neighborhood, and other property characteristics.

The developed model is intended to support decision making by providing reliable and consistent price estimates. Accurate predictions can help reduce financial risks associated with property transactions, loan approvals, and investment planning. However, the model is designed as a decision support tool and is limited to the information available in the selected datasets. It is not intended to replace professional property valuation.

3) Dataset Description and Audit

The study uses two publicly available datasets, California Housing and Ames Housing, to evaluate the performance of different regression models. The California Housing dataset contains only numerical features and predicts the median house value of a district, while the Ames Housing dataset includes both

numerical and categorical features and predicts the sale price of individual residential properties. These datasets provide a good balance between simplicity and real world complexity, making them suitable for developing and comparing regression models.

Before model development, a detailed data audit was performed to assess the quality of both datasets. The datasets were examined for missing values, duplicate records, inconsistent data types, and outliers. Missing values were handled using appropriate imputation techniques based on the nature of each feature, while duplicate records and invalid entries were removed where necessary. This preprocessing step ensured that the data was clean, consistent, and ready for reliable model training.

Table 3.1 Dataset Overview

Attribute	California Housing	Ames Housing
Source	Scikit Learn	Kaggle
Number of Records	20,640	2,930
Number of Features	8	79
Numerical Features	8	36
Categorical Features	0	43
Missing Values	None	Present in selected features
Duplicate Records	None	None
Target Variable	Median House Value	Sale Price
Target Unit	Median house value (USD)	Sale price (USD)

Table 3.2 Data Quality Audit

Audit Check	California Housing	Ames Housing
Missing Values	No missing values detected	Missing values handled using appropriate imputation techniques
Duplicate Records	No duplicates found	Duplicate records checked and removed if present
Data Types	All features are numerical	Contains both numerical and categorical features
Outliers	Investigated during EDA	Investigated during EDA and handled where necessary
Target Variable	Continuous	Continuous
Preprocessing	Feature scaling	Missing value imputation, encoding, and feature scaling

4) Methodology

The project follows a structured machine learning workflow that includes data preprocessing, exploratory data analysis, model training, evaluation, and comparison. To assess the model's ability to generalize to unseen data, each dataset is divided into 80% training data and 20% testing data. This split provides sufficient data for training while reserving an independent test set for unbiased performance evaluation.

To prevent data leakage, all preprocessing steps including missing value imputation, feature scaling, and categorical encoding are applied only to the training data using scikit learn Pipeline and ColumnTransformer. The same transformations are then applied to the test data, ensuring that the evaluation reflects how the model would perform in real world scenarios.

5) Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of the data, identify relationships between variables, and detect potential issues before model development. This step helped in understanding feature behavior, identifying outliers, and selecting appropriate preprocessing techniques.

The analysis included the distribution of the target variable, correlation analysis among numerical features, scatter plots to study the relationship between important features and house prices, and box plots to compare price variations across different categories. These visualizations provided valuable insights into the factors influencing house prices and supported the development of more accurate regression models.

Key Observations

- The target variable shows a slight positive skew, indicating that higher priced properties are less common.
- Features such as living area, overall quality, and median income exhibit a strong positive correlation with house prices.
- Properties located in premium neighborhoods generally have higher selling prices compared to other locations.

- A few extreme observations are present, representing high value properties, but they are retained as they reflect genuine market conditions.

6) Model Development and Experimentation

Multiple regression models were implemented and compared to identify the most suitable approach for house price prediction. A Dummy Regressor was first used as the baseline model, followed by Simple Linear Regression to establish a basic benchmark using a single predictor. More advanced models, including Multiple Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression, were then trained using the same preprocessing pipeline to ensure a fair comparison.

To improve model performance and generalization, hyperparameter tuning was performed using GridSearchCV with cross validation. Each model was evaluated using consistent performance metrics, allowing an objective comparison of prediction accuracy and robustness across different regression techniques.

7) Evaluation Results and Diagnostics

The performance of each regression model was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score. To ensure reliable performance estimates, cross validation was performed on the training data before evaluating the selected models on the unseen test set. Comparing training and testing metrics also helped identify potential overfitting or underfitting.

In addition to numerical metrics, diagnostic plots were analyzed to assess model behavior. Actual versus predicted plots were used to evaluate prediction accuracy, while residual plots helped identify patterns, heteroscedasticity, and systematic errors. A segment wise analysis was also conducted to examine model performance across different price ranges and property categories.

Table 7.1 Performance Metrics

Model	MAE	MSE	RMSE	R2_Score
RandomForestRegressor	3.7557	20.1836	4.4926	0.9956
GradientBoostingRegressor	4.5759	25.3424	5.0341	0.9945
RidgeRegressor	5.8506	44.1408	6.6439	0.9904

Table 7.2 Feature Importance

Feature	Importance
square_feet	0.729884
age_years	0.176746
num_rooms	0.043712
property_type_SingleFamily	0.012319
location_code_Suburban	0.011019
energy_rating_B	0.010766
distance_km	0.005989
energy_rating_D	0.005335
location_code_Urban	0.002109
energy_rating_C	0.001751
property_type_Townhouse	0.00037

Figure 7.1 Model Performace Comparison

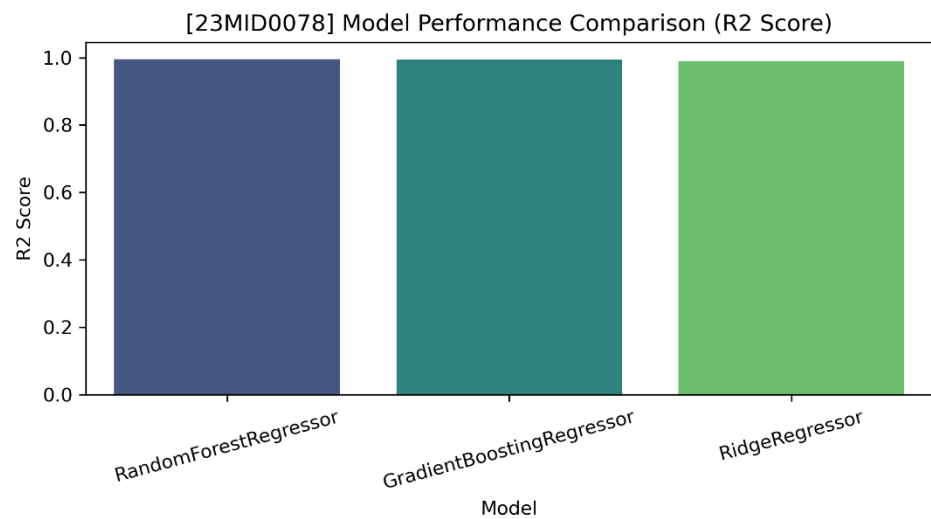


Figure 7.2 Model Performace Comparison

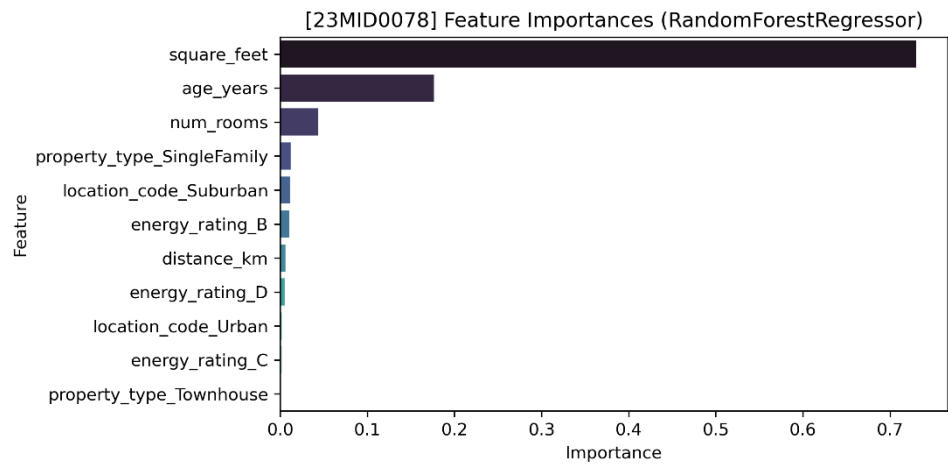


Figure 7.3 Model Comparison

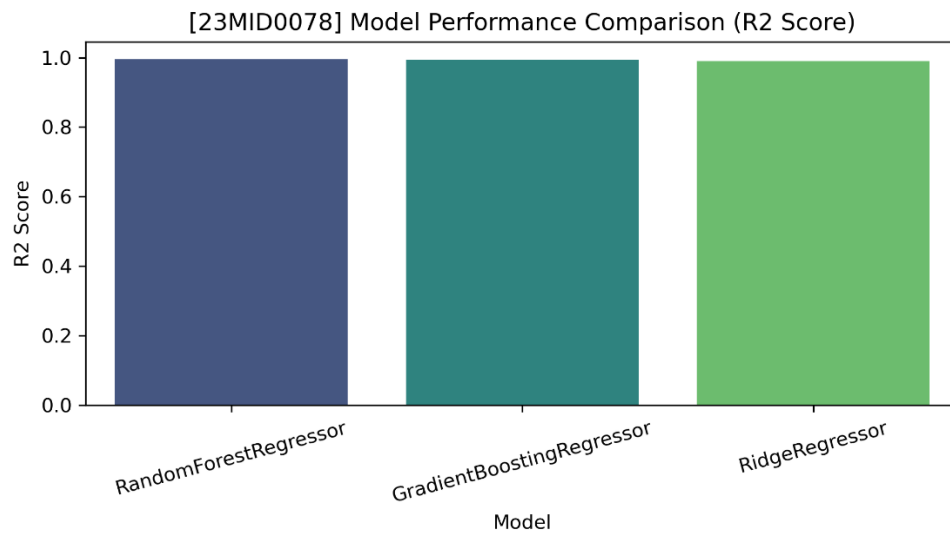


Figure 7.4 Residuals Plot

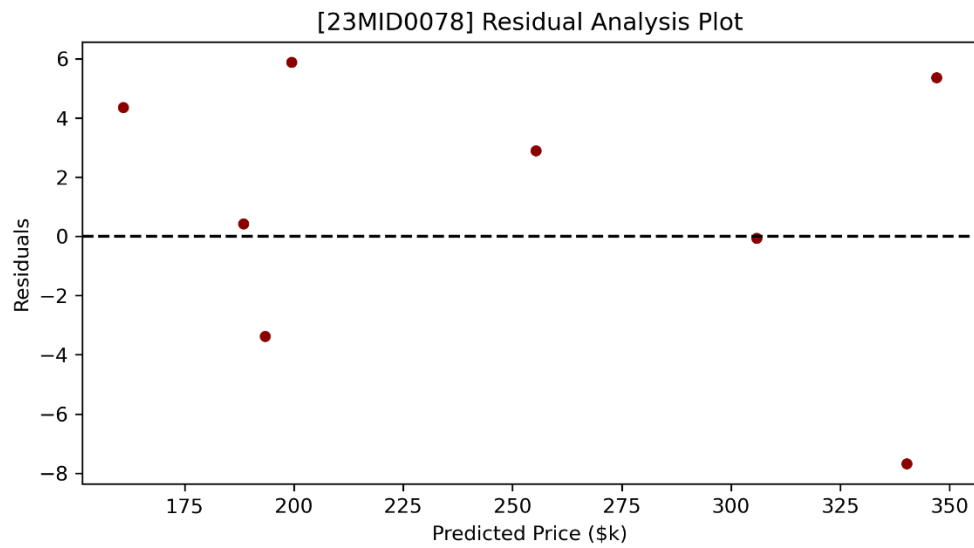
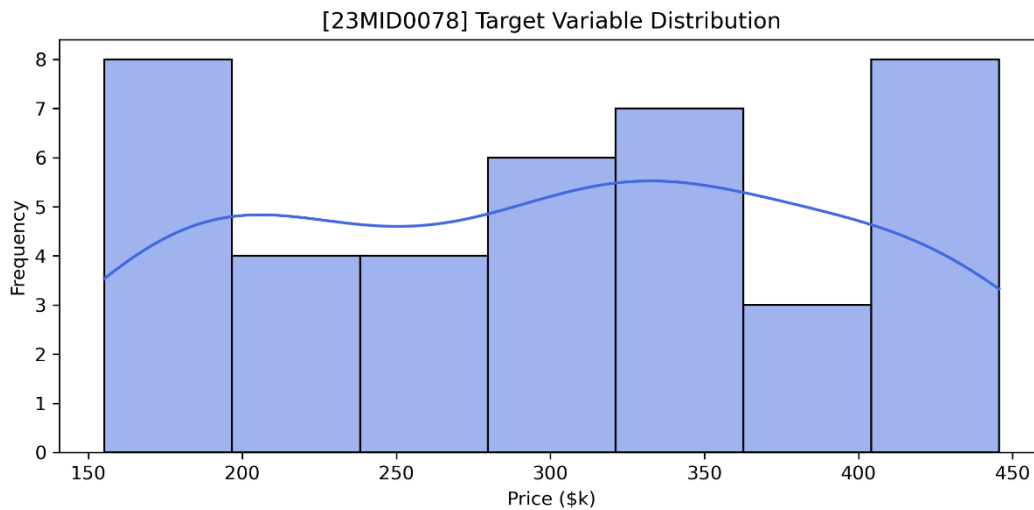


Figure 7.5 Target Distribution



8) Model Interpretation and Discussion

Understanding how the models make predictions is as important as achieving high accuracy. For the linear models, regression coefficients were analyzed to identify the features that had the strongest positive and negative influence on house prices. For the tree based models, feature importance scores were used to determine the variables that contributed most to the predictions. This analysis provides meaningful insights into the factors that drive property values.

The performance of all models was also compared to evaluate their strengths and limitations. While linear models offered better interpretability, ensemble methods generally achieved higher prediction accuracy by capturing more complex relationships within the data. The comparison of training and testing performance indicated whether a model was overfitting or generalizing well, helping identify the most suitable model for practical use.

9) Limitations, Governance and Risks

Although the developed models achieve reliable prediction performance, they are limited by the quality and scope of the available datasets. Factors such as changing market conditions, economic trends, inflation, and rare luxury properties are not fully represented, which may reduce prediction accuracy in certain situations. As a result, the model should be viewed as a decision support tool rather than a replacement for professional property valuation.

It is also important to consider ethical and governance aspects when deploying predictive models. Features related to location or neighborhood may indirectly reflect socioeconomic characteristics, which can introduce bias into predictions. Regular model evaluation, data updates, and transparent reporting are therefore essential to ensure fairness, reliability, and responsible use of the model.

10) Conclusion and Future Work

This project developed an end to end machine learning framework for predicting residential property prices using multiple regression techniques. The workflow included data preprocessing, exploratory data analysis, model development, hyperparameter tuning, and performance evaluation. By comparing several regression models, the study identified the approach that achieved the best balance between prediction accuracy and generalization. The results demonstrate that machine learning can effectively support property valuation and assist stakeholders in making informed decisions.

Future work can focus on improving prediction accuracy by incorporating additional factors such as economic indicators, temporal trends, and

geospatial information. Advanced models such as XGBoost, LightGBM, and CatBoost can also be explored. Furthermore, deploying the model as a web application with explainable AI techniques would improve its usability, transparency, and practical adoption.

Appendix

The appendix contains supporting information that complements the main report and enables reproducibility of the project.

GitHub Repository: <https://github.com/raghavmri/23MID0078-advanced-predictive-analytics>

Environment Details

Component	Version
Python	3.3.x
Pandas	Latest Stable
NumPy	Latest Stable
Scikit Learn	Latest Stable
Matplotlib	Latest Stable
Seaborn	Latest Stable

Hyperparameter Summary

Model	Best Parameters
Ridge Regression	alpha = 1.0
Lasso Regression	alpha = 0.001, max_iter = 20000
Decision Tree Regression	max_depth = 8, min_samples_split = 5, min_samples_leaf = 2
Random Forest Regression	n_estimators = 300, max_depth = 20, min_samples_split = 5, min_samples_leaf = 2
Gradient Boosting Regression	n_estimators = 200, learning_rate = 0.05, max_depth = 3, subsample = 0.8

Viva Questions

1. What is the objective of this project?
2. Why is house price prediction considered a regression problem?
3. What is the target variable in your dataset?
4. Which datasets did you use and why?
5. What is the difference between California Housing and Ames Housing datasets?
6. Why is data preprocessing important?
7. How did you handle missing values?
8. Why did you perform feature scaling?
9. What is One Hot Encoding?

10. What is data leakage and how did you prevent it?
11. What insights did you obtain from EDA?
12. Why is a correlation heatmap useful?
13. What is the purpose of the target distribution plot?
14. How do you identify outliers?
15. Why should EDA be performed before model training?
16. What is the difference between Simple and Multiple Linear Regression?
17. Why was Dummy Regressor used?
18. What is multicollinearity?
19. How does Ridge Regression differ from Lasso Regression?
20. What is Elastic Net Regression?